



Basic Electrical and Electronics Engineering

This practicum introduces electrical quantities, DC and AC circuits, safety, measuring instruments, passive components, semiconductor devices, rectification and logic gates. Theory is tied directly to breadboard, meter, DSO and safety-device activities.

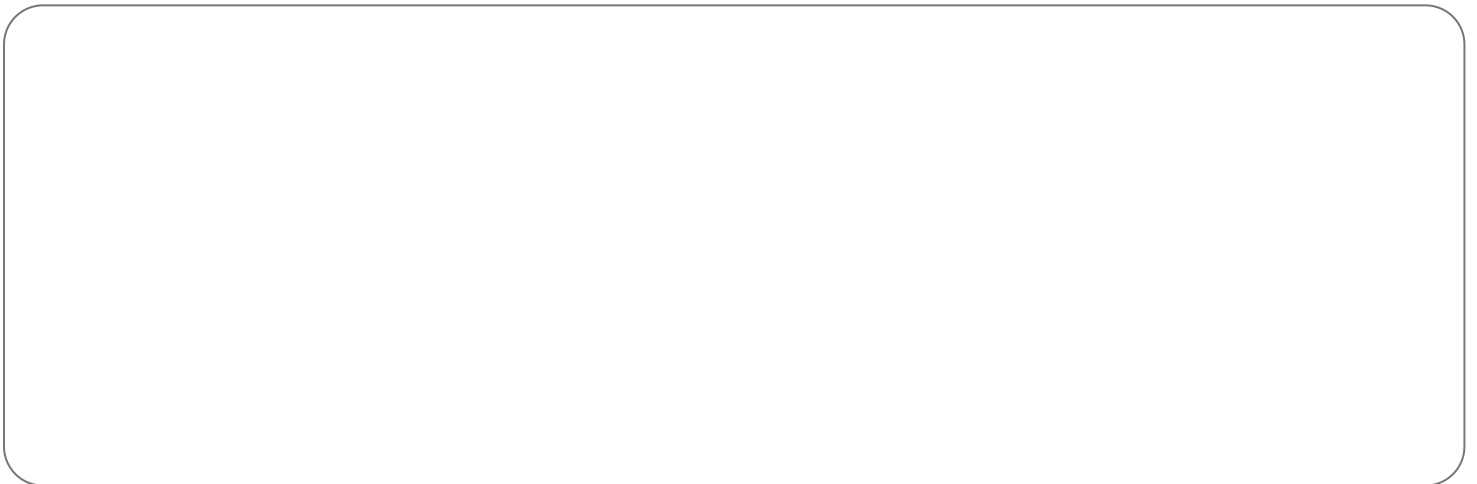
REV2026

Semester I

Practicum

Malayalam support

Printable handbook



COURSE OVERVIEW

What this handbook covers

The structure follows the supplied official SITTTTR Kerala Revision 2026 syllabus. It is a learning handbook rather than a replacement for the official syllabus PDF. Use the module checklist to ensure that every prescribed point has been revised.

CO1

Apply fundamental electrical quantities, Ohm's law, resistor combinations and division rules.

CO2

Interpret AC parameters, electrical safety devices and power in single-phase circuits.

CO3

Identify, test and apply resistors, capacitors and inductors using appropriate instruments.

CO4

Explain and demonstrate semiconductor devices, half-wave rectification, transistors and logic gates.

Module roadmap

Module I

Electrical fundamentals, resistor networks and measurement

Module II

AC parameters, electrical safety and AC power

Module III

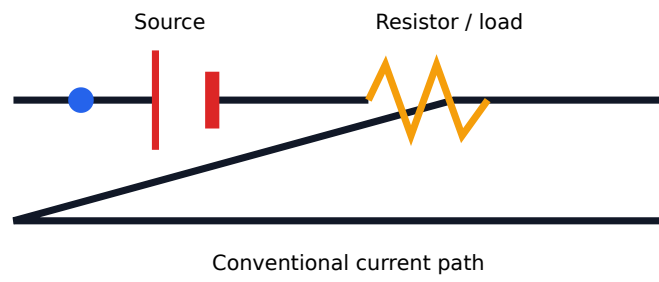
Passive components, breadboard, DSO and measurement

Module IV

Semiconductors, diode, rectifier, transistor and logic gates

VISUAL CONCEPT

See the subject as a system



Electrical fundamentals, resistor networks and measurement

20 (11 L + 9 P)

CO1

Voltage potential difference V ; Current charge flow rate I ; Resistance current- I oppose V . Ammeter series- I voltmeter parallel- V connect V . Range V meter connect I instrument damage V .

Electrical quantities and units

Define charge, current, voltage, resistance, power and energy. Use A, V, Ω , W, J, Wh and kWh correctly and distinguish power from energy.

Series resistance

Same current flows through every resistor; equivalent resistance is the sum. Voltage divides in proportion to resistance.

Voltage and current division

Use $V_x = V_s R_x / \sum R$ for series networks and branch-current relations for parallel networks. Confirm results with KVL/KCL.

DC voltage/current measurement

Connect a voltmeter across the element and an ammeter in series. Apply the correct multiplication factor on a multirange instrument.

Ohm's law

For a conductor at constant physical conditions, $V = IR$. Solve for any unknown and check whether the assumed linear relationship is valid.

Parallel resistance

Same voltage appears across branches; branch currents add. Equivalent resistance is less than the smallest branch resistance.

Symbols and instruments

Identify source, switch, resistor, lamp, fuse, earth, ammeter, voltmeter, wattmeter and rheostat symbols. Select instrument, function, AC/DC mode and range.

Rules / common mistakes

- De-energise before changing wiring.
- Start with the highest meter range.
- Never place an ammeter directly across a source.
- Check polarity for DC measurements.

Practice and self-learning

- Identify basic circuit symbols and measuring instruments.
- Use tester and multimeter safely.
- Build a DC resistor circuit and measure voltage/current; compare with calculation.

AC parameters, electrical safety and AC power

20 (11 L + 9 P)

CO2

230 V household supply RMS value 230 V ; peak value $230\sqrt{2}\text{ V}$. MCB overcurrent/short circuit protection; RCCB leakage current detect shock risk function.

Sinusoidal terms

Define cycle, frequency, time period, angular frequency, amplitude, instantaneous value, maximum value, RMS value, average value, form factor and peak factor.

Instantaneous equation

Write $v = V_m \sin(\omega t + \phi)$, where $\omega = 2\pi f$. Relate frequency and period by $T = 1/f$.

RMS and average values

For a sine wave $V_{rms} = V_m/\sqrt{2}$, V_{avg} over a half-cycle $= 2V_m/\pi$, form factor ≈ 1.11 and peak factor $= \sqrt{2}$.

Electrical safety precautions

Isolation, earthing, dry hands, rated tools, lockout/tagout where applicable, correct PPE, one-hand technique for live diagnostics and immediate first-aid response.

Fuse, MCB, MCCB, ELCB and RCCB

Explain operating purpose, rating and limitation. A fuse/MCB responds to overcurrent; RCCB responds to residual current and does not replace overcurrent protection.

AC power

Define active $P = VI \cos\phi$, reactive $Q = VI \sin\phi$ and apparent $S = VI$. Explain power factor and the power triangle.

Measurement of AC parameters

Construct a simple source-switch-fuse-load circuit and measure voltage, current and power using correctly rated instruments.

Rules / common mistakes

- Treat every conductor as live until proved dead.
- RCCB test button should be checked periodically.
- Do not bypass a protective device.
- Use instrument category and voltage rating appropriate to the installation.

Practice and self-learning

- Identify safety devices in a residential distribution board.
- Sketch a 230 V, 50 Hz waveform and calculate peak/time period.
- Measure voltage, current and power of a safe laboratory AC load.

Passive components, breadboard, DSO and measurement

23 (11 L + 12 P)

C03

Resistor value colour bands $\square\square\square\square\square\square$; capacitor- $\square$ polarity/voltage rating $\square\square\square\square\square\square\square\square$; inductor current change oppose $\square\square\square\square\square\square\square$. DSO waveform $\square\square\square\square\square\square\square\square\square\square\square\square\square\square$ voltage/div, time/div, probe factor $\square\square\square\square\square$ $\square\square\square\square\square\square\square$ actual value $\square\square\square\square\square\square\square\square$.

Resistors

Fixed carbon composition and wire-wound resistors; variable potentiometer, rheostat and trimmer. Select resistance, tolerance and power rating.

Inductance

Define inductance, self-inductance and mutual inductance, measured in henry. Explain energy storage in magnetic field and common applications.

Breadboard construction

Recognise power rails and connected terminal groups. Build a 3 V LED circuit from a 5 V source using a current-limiting resistor and switch.

LCR measurement

Identify capacitor types/codes, measure capacitance and inductance, and construct a simple wound inductor on a toroidal or EE core without design calculation.

Four-band colour code

First two bands are significant digits, third is multiplier and fourth is tolerance. Verify decoded value with a digital multimeter.

Capacitance

Define capacitance, fixed and variable capacitor types, polarity and voltage rating. Use series and parallel combination equations.

DSO operation

Identify vertical/horizontal controls, select $1\times/10\times$ probe, compensate probe with calibration output, set trigger and compute $V_{\max}$, V_{pp} and period.

Rules / common mistakes

- Discharge capacitors before handling or measuring.
- Observe electrolytic capacitor polarity.
- Include probe attenuation in DSO calculations.
- Use a current-limiting resistor with an LED.

Practice and self-learning

- Construct an LED circuit on a breadboard.
- Operate a DSO and measure amplitude and period.
- Decode and verify resistor/capacitor values; measure a wound inductor.

Semiconductors, diode, rectifier, transistor and logic gates

21 (12 L + 9 P)

C04

PN junction forward bias- barrier potential current flow depletion region; reverse bias- depletion region. Diode one-way device. Transistor terminals logic-gate truth table.

Active and passive components

Passive parts do not provide power gain; active semiconductor devices can control current or provide gain when supplied correctly.

PN junction diode

Construction, depletion region, barrier potential, forward and reverse bias, polarity identification and basic applications.

BJT basics

Identify NPN and PNP symbols, emitter, base and collector. Explain transistor action qualitatively and common uses as switch or amplifier.

Device identification and verification

Use marking, package orientation and meter diode mode to identify diode/transistor terminals; verify logic truth tables safely.

Intrinsic and extrinsic semiconductors

Explain doping, N-type majority electrons, P-type majority holes and the role of minority carriers.

Half-wave rectifier

One diode conducts during one half-cycle. Explain input/output waveforms, peak inverse voltage, ripple and the effect of a filter capacitor.

Logic gates

Symbols and truth tables of AND, OR, NOT, NAND, NOR, XOR and XNOR. Explain why NAND and NOR are universal gates.

Rules / common mistakes

- Use a series resistor with LEDs and diode tests.
- Do not exceed diode reverse rating or transistor current rating.
- Tie unused logic inputs to a defined level.
- Switch off supply before changing breadboard connections.

Practice and self-learning

- Identify diode and transistor terminals.
- Construct half-wave rectifier with and without filter; plot waveforms.
- Verify AND, OR, NOT, NAND, NOR and XOR truth tables.

QUICK REVISION

Formula / rule bank

Rule / Formula 1

$$I = Q/t$$

Rule / Formula 2

$$V = IR$$

Rule / Formula 3

$$P = VI = I^2R = V^2/R$$

Rule / Formula 4

$$E = Pt; 1 \text{ kWh} = 3.6 \text{ MJ}$$

Rule / Formula 5

$$\text{Series R: } R_{eq} = R_1 + R_2 + \dots$$

Rule / Formula 6

$$\text{Parallel R: } 1/R_{eq} = \Sigma(1/R)$$

Rule / Formula 7

$$\text{Voltage divider: } V_x = V_s R_x / \Sigma R$$

Rule / Formula 8

$$T = 1/f; \omega = 2\pi f$$

Rule / Formula 9

$$V_{rms} = V_m / \sqrt{2}; V_{avg}(\text{half-cycle}) = 2V_m / \pi$$

Rule / Formula 10

$$P = VI \cos\phi; Q = VI \sin\phi; S = VI$$

Rule / Formula 11

$$\text{Parallel C: } C_{eq} = \Sigma C; \text{ Series C: } 1/C_{eq} = \Sigma(1/C)$$

Rule / Formula 12

$$\text{LED resistor: } R = (V_s - V_f) / I$$

Worked checkpoints

Worked checkpoint 1

A 12 V source across 6 Ω gives $I = 2$ A and $P = 24$ W.

Worked checkpoint 2

For 230 V RMS sine, $V_m \approx 325$ V and $T = 20$ ms at 50 Hz.

Worked checkpoint 3

Resistors 2 k Ω and 3 k Ω in series across 10 V: current = 2 mA and voltage drops are 4 V and 6 V.

Worked checkpoint 4

A 5 V source, LED V_f 2 V at 10 mA requires about 300 Ω ; choose the next safe standard value such as 330 Ω .

APPLICATION

Practical, drawing and self-learning work

No.	Task	Required evidence
1	Circuit symbols and instrument identification.	Record procedure, observation/result and safety/quality checks.
2	Tester/multimeter operation and DC measurement.	Record procedure, observation/result and safety/quality checks.
3	Electrical safety devices and precautions.	Record procedure, observation/result and safety/quality checks.
4	AC voltage/current/power measurement.	Record procedure, observation/result and safety/quality checks.
5	Breadboard LED circuit.	Record procedure, observation/result and safety/quality checks.
6	DSO operation and waveform measurement.	Record procedure, observation/result and safety/quality checks.
7	Resistor colour-code verification.	Record procedure, observation/result and safety/quality checks.
8	Capacitor identification and LCR measurement.	Record procedure, observation/result and safety/quality checks.
9	Wound inductor and inductance measurement.	Record procedure, observation/result and safety/quality checks.
10	Diode/transistor identification.	Record procedure, observation/result and safety/quality checks.
11	Half-wave rectifier with/without filter.	Record procedure, observation/result and safety/quality checks.
12	Logic-gate truth-table verification.	Record procedure, observation/result and safety/quality checks.

EXAM PREPARATION

Expected questions and model-answer guides

These are syllabus-aligned practice questions, not a claim about the exact end-semester paper.

2 marks

Q1. Define current, voltage, resistance, power and energy with units.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q2. State Ohm's law and solve a simple circuit.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q3. Derive equivalent resistance for series and parallel combinations.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q4. Explain voltage and current division rules.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q5. Describe correct connection of ammeter and voltmeter.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q6. Define AC cycle, frequency, time period, RMS and average value.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q7. Compare fuse, MCB and RCCB.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q8. Explain active, reactive and apparent power.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q9. Decode a four-band resistor and verify it using a meter.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q10. Explain DSO voltage/div, time/div and probe factor.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q11. Explain PN junction formation and biasing.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q12. Draw and explain a half-wave rectifier.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q13. Explain NPN/PNP transistor symbols and action.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q14. Write truth tables for basic and universal gates.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

COVERAGE AUDIT

Official syllabus coverage checklist

- Module I: Electrical quantities and units
- Module I: Ohm's law
- Module I: Series resistance
- Module I: Parallel resistance
- Module I: Voltage and current division
- Module I: Symbols and instruments
- Module I: DC voltage/current measurement
- Module II: Sinusoidal terms
- Module II: Instantaneous equation
- Module II: RMS and average values
- Module II: Electrical safety precautions
- Module II: Fuse, MCB, MCCB, ELCB and RCCB
- Module II: AC power
- Module II: Measurement of AC parameters
- Module III: Resistors
- Module III: Four-band colour code
- Module III: Inductance
- Module III: Capacitance
- Module III: Breadboard construction
- Module III: DSO operation
- Module III: LCR measurement
- Module IV: Active and passive components
- Module IV: Intrinsic and extrinsic semiconductors
- Module IV: PN junction diode
- Module IV: Half-wave rectifier
- Module IV: BJT basics
- Module IV: Logic gates
- Module IV: Device identification and verification

REFERENCES

Learning resources

1. A Textbook of Electrical Technology Part I — B. L. Theraja & A. K. Theraja
2. Basic Electronics and Linear Circuits — N. N. Bhargava et al.
3. Basic Electrical Engineering — V. K. Mehta & Rohit Mehta
4. Electronic Devices and Circuits — G. K. Mithal

Primary authority for scope: supplied official SITTTR Kerala Diploma Curriculum Revision 2026 syllabus PDF for Course 1031.